

NoSQL Databases

NoSQL (Not Only SQL)

RDMS:relational

Limitations:

Not good for big data

Not a structured data

No sql deals with these limitations ,they handle bigdata

Here store value as

Key value:redids,Riak

Columnar: Hbase,cassandra

Document db :mongo db

Graphical :NE04

Relational vs Document Database

Feature	RDMS	DOCUMENT DATABASE
Structure	table	collection
Relation	foreign key,join	flexible/dymanic schema
Scaling	vertical	horizontal
Examples	mysql	Mongodb

Key difference

RDBMS stores structured tabular data

Mongodb stores data as BSON (BINARY JSON)documents.

In vertical the scaling is done in 1 machine itself like adding ram to the lap

In horizontal scaling we will be using clusters as we will be adding more machines to network

Mongodb Structure

Mongodb hierarchy:

Server

->database

->collections

->document(json)

Example:

```
{
  "Id":1,
  "Name": "anvin",
  "Department": "it",
  "Salary":75000
}
```

Create db

Use EmployeeDB

```
db.createCollection('employee')
```

```
db.employees.insertOne({  
  Name:"anvi",  
  salary:70000})
```

```
db.employees.insertMany([  
  {name:"anvi",salary:30000},  
  {name:"sam",salary:35000}  
])
```

```
db.employees.updateOne(  
  db.employees.updateMany(  
    db.employees.deleteOne({name:"sam"})  
    db.employees.find()  
    db.employees.find({department:"IT"})  
    db.employees.aggregate([  
      {$group: {_id:"$department",avgSalary:{$avg:"salary"}}}  
    ])  
    db.employees.aggregate([  
      {$group: {_id:"$department",totalEmployees:{$sum:1}}}  
    ])  
    db.employees.aggregate([  
      {$sort:{salary:-1}}  
    ])
```

Operator and their description

\$set=update field value

\$inc=increment value

\$push=add element to array

\$unset=remove field

Delete one

```
db.employees.deleteOne({name:"Rahul"})
```

Delete many

```
db.employees.deleteMany({department:"finance"})
```

```
db.employees.createIndex({department:1})
```

```
db.createIndex({department:1,salary:-1})
```

```
db.employees.getIndexes()
```